Venkata Lakshmi Parimala Pasupuleti

Agentic AI Engineer · LangGraph · Multi-Agent Systems · RAG · Tool Orchestration · LLM Evaluation · Housing & Hospitality Domain · 9yr Production

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PROFESSIONAL SUMMARY

I built LeaseI -- an 8-agent LangGraph agentic platform for real estate compliance workflows. The architecture: a router that classifies intent, an orchestrator that assembles context from a complex regulatory corpus, tool calls that retrieve and validate against live rules, a synthesizer that generates responses, and a HITL escalation gate when confidence drops below threshold. That is Conduit's platform applied to housing and hospitality -- I built it because the same coordination problem exists in leasing and property management. I also built AgentProbe: a CI-integrated eval framework with LLM-as-judge, multi-trial sampling, hallucination detection, and a regression gate that blocks deploys when correctness scores degrade. I don't ship a prompt change without knowing what it does across 1,240 conversations. On the domain: I spent 8 years at Entrata building leasing, onboarding, scheduling, and retention workflows for rental housing -- the same operations Conduit automates for hospitality. I bring the agentic platform (built it), the eval discipline (built that too), and the domain fluency (8 years). Portfolio: venkatalakshmiparimala.github.io/Portfolio/about.html

TECHNICAL SKILLS

Agentic AI & Evaluation: LangGraph (multi-agent orchestration), Agentic AI, Multi-Agent Systems, RAG Pipelines, Tool Orchestration, Function Calling, LLM Evaluation, LLM-as-Judge, Observability, Agent Tracing, Hallucination Detection, Confidence Scoring, HITL, Guardrails, Response Pipeline Design, Retrieval Reliability, Document Intelligence, OCR, Embeddings, Semantic Search, LangChain, Prompt Engineering, Production AI Systems, Claude Code (daily tool)

Languages & Frameworks: Python (9yr production), TypeScript, JavaScript, SQL/PL-SQL, R, Bash/shell scripting, HTML/CSS; FastAPI, Flask, Angular, React.js; LLM APIs (Anthropic, OpenAI, Groq, Ollama)

Backend, Data & Infrastructure: PostgreSQL (200+ queries optimized, stored procedures, multi-tenant schemas), MySQL, Oracle, NoSQL, ETL/ELT Pipelines, Data Modeling, REST APIs, Microservices, AWS (S3, EC2, RDS, Glue), Docker, Kubernetes, RabbitMQ async processing, CI/CD (GitHub Actions), Automated Testing, Regression Testing, Git, Agile/Scrum

EXPERIENCE

Entrata -- Multifamily Real Estate SaaS (PropTech)

Sep 2018 -- Aug 2024

Senior Software Engineer -- Leasing, Onboarding, Scheduling & Retention Workflows · Lehi, UT (Remote)

- **8 years building the workflows Conduit automates for hospitality** -- leasing (HUD/LIHTC eligibility, income certification, lease compliance), onboarding (document management, move-in workflows, resident portals), scheduling (rent-schedule updates, certification cycles, recertification timelines), and retention (compliance renewals, HAP calculations, occupancy data) for 100K+ housing units. Zero compliance violations across full ownership period. The guest communication and operations coordination problems Conduit is solving in hospitality are structurally identical to what I built for housing.
- **Full stack production delivery** -- shipped 20+ full-stack modules (Python, Angular/JavaScript, PostgreSQL) across 120+ Agile sprints; Docker microservices redesign (50% scalability improvement, 30% deployment time reduction); eliminated 80% of manual operations workload through Python/Bash automation pipelines. 300+ peer code reviews; mentored 5+ engineers.
- **Data reliability at scale** -- RabbitMQ async processing with idempotency guarantees; optimized 200+ SQL queries (30% performance gain); complex multi-tenant schemas; AWS S3 migration with Libsodium-encrypted ETL pipelines (100% PHI compliance, zero security incidents).

George Mason University

Aug 2024 -- May 2026

Research Associate & Graduate Teaching Assistant

- Built multi-source ML prediction pipelines (Python, R) across 190+ countries (R^2=0.977); applied Isolation Forest to automate data integrity checks across 3 research pipelines; designed Tableau and ggplot2 dashboards improving research comprehension 40% for faculty leadership; mentored 600+ students in Python, SQL, and data engineering.

Magizhs Tech Solution

May 2025 -- Aug 2025

AI/ML Engineer -- Agentic Workflow Orchestration & Regulatory RAG

- Built an agentic system (LangGraph stateful graph: parse → plan → retrieve → validate → conditional routing) that retrieves from a regulatory corpus (FAA Part 107), validates against rules, and decides whether to proceed or escalate -- the same orchestration pattern as Conduit's guest workflow agents routing between response, escalation, and tool-call paths. Guardrails enforce compliance rules before export; every decision grounded in cited regulation.

Entrata India -- Pune

Jan 2017 -- Sep 2018

Associate Software Engineer -- Python · PostgreSQL · Housing Compliance Workflows

- Built property management and compliance modules (Python, PostgreSQL, Angular); income-limit import pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints; 56 state-specific housing certification templates.

PROJECTS

Leasel: 8-Agent Agentic Platform -- Router, Orchestrator, Synthesizer, Escalation

LangGraph · Agentic AI · RAG · Tool Orchestration · HITL · LLM Evaluation

Python · LangGraph (8-agent StateGraph) · FastAPI · TF-IDF RAG · SQLite · OCR · pdfplumber · GitHub Actions | github.com/venkatalakshmiparimala/Lease_complaince_rag

Agentic platform for housing compliance -- the architecture Conduit applies to hospitality guest communication. **The pipeline:** a Triage agent classifies intent and routes to the appropriate branch; Retrieval handles knowledge retrieval from a complex regulatory corpus (TF-IDF RAG, section-aware chunking, missing/conflicting clause detection, 100% citation correctness); Configuration and Historical agents add live system state and incident context; Reasoning generates the response; Validation scores confidence; HITL Escalation fires when confidence drops below threshold -- never auto-resolves regulated decisions. Same router → orchestrator → synthesizer → escalate pattern as Conduit's guest workflow agents. **Results:** 93.6% precision and 87.9% F1 across 1,240 compliance test cases (5 US states + federal, 1,050 synthetic cases); pass@1 81.7%, pass@3 88.3%; CI regression gate blocking deploys on score degradation. **Deliberate model choice:** TF-IDF RAG over a vector database -- more reliable for compliance retrieval where exact clause matching matters more than semantic proximity.

AgentProbe: CI-Integrated Eval Framework -- Regression Gate for Agentic Systems

LLM Evaluation · Observability · CI/CD · Multi-Provider · FastAPI · React

Python · FastAPI · React.js · TypeScript · ChromaDB · GitHub Actions · Anthropic / OpenAI / Groq APIs | github.com/venkatalakshmiparimala/AgentProbe

I built AgentProbe because I needed an evaluation framework to trust LeaseI in production. 5-metric automated evaluation engine (correctness, groundedness, latency, token efficiency, hallucination rate); CI regression gate (GitHub Actions) that blocks deploys when quality scores degrade; structured observability -- what was retrieved, what confidence was, where the agent branched; multi-provider LLM routing (Anthropic, OpenAI, Groq); LLM-as-judge scoring with multi-trial sampling; 15-endpoint FastAPI backend + React/TypeScript frontend. I don't ship a prompt change without knowing what it does across 1,240 conversations. That discipline is why AgentProbe exists -- and it's what Conduit's eval and observability requirements describe.

AI on FHIR: NLP-Driven Full Stack Query Platform

Flask · React · TypeScript · LLMs · Full Stack · HIPAA

Python · Flask · React · Docker · spaCy · BERT · Embeddings · FHIR API | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask + React full-stack platform translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings -- reduced manual retrieval time ~60%; HIPAA-compliant RBAC across structured medical records. Full-stack delivery proof: backend API design, frontend React/TypeScript, containerized deployment.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2013--2017